

SPACE MISSIONS

Setback in Space

At 2.57 a.m. on September 6, while the nation slept, a bleary-eyed Promod Kale, project director of the Indian National Satellite System INSAT-1A, depressed three switches at the Master Control Facility at Hassan, Karnataka. The final switch snapped contact with the crippled gold and blue satellite, which lurched drunkenly 36,000 km above the ground, west of Delhi. "Our bird is dead," mourned a space scientist in Bangalore.



LANDMARK ODYSSEY: Rakesh Sharma (right) with Indira Gandhi before his 1984 space journey

The multi-purpose Rs 60-crore satellite, providing a unique combination of telecommunication, meteorological and mass communication capabilities, lasted barely 148 days after it was launched from Cape Canaveral on April 10. With its premature death, India's giant leap into an era of operational space systems went into an unaccountable tailspin. An Indian Space Research Organisation (ISRO) scientist admitted, "It's a shame for all of us. Our plans have suffered a serious setback." With INSAT-1A dead, over Rs 150 crore worth of related ground facilities now wait for a second satellite.

Raj Chengappa, September 30, 1982, INDIA TODAY

moon or the planets or manned space flight." Today India's space programme is one of the best in the world with an annual budget of \$660 million (Rs 2,640 crore), though this is just three per cent of NASA's space budget.

When the first sounding rocket was launched on November 21, 1963, from Thumba, everything was from abroad except for the Indian engineers and the launch station. The 715 kg-rocket was an American-built Nike Apache, the sodium vapour payload was French and the range clearance was given by the Soviet Union. But 44 years later, India has put in orbit satellites for seven other countries, including Germany and Italy. The programme has grown exponentially as the first launch vehicle SLV-

3, with a modest payload capability of 40 kg, has grown to the soon-to-be-ready GSLV Mark III which can carry satellites weighing more than 4,000 kg. India also has two of the world's largest networks of communication and remote sensing satellites—the Indian National Satellite (INSAT), that facilitates telecommunication, television broadcasting, meteorology, disaster warning, search and rescue and the Indian Remote Sensing (IRS) satellite system, that offers

space-based data for applications in agriculture, water resources, urban development and forestry.

Though understated, India's space programme has radically impacted the life of common man. Satellites have taken television to 90 per cent of India, provided daily weather forecasts and facilitated educational and tele-medicine projects. Images from remote sensing satellites have helped farmers increase crop yields and the fishermen increase catch. Heavy casualties were averted in the super cyclone of 1999 in Orissa because of the early warning given by satellites. Satellite-aided payloads receive signals from distress beacons in sea, land and water and a tele-medicine project links 165 hospitals in villages with 33 specialty hospitals in cities enabling more than 15,000 tele-consultations for villagers every year. Any criticism over the viability of a developing country like India pursuing a fancy space programme has long been silenced. ■